

AP CSP | Lesson 21: Robot Algorithms

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: A

A follows a 90-degree counterclockwise turn. B is a right turn. C ignores rotation. D is a half turn. Position does not change.

Q2. Answer: B

B moves one column east. A assumes forward always means north. C moves south. D ignores a valid movement.

Q3. Answer: C

C separates sensing from action. A and B add movement effects absent from the reference. D confuses a Boolean result and an absolute compass direction.

Q4. Answer: D

D matches the official rule. A and B invent automatic navigation. C omits termination and would produce incorrect traces of later commands.

Q5. Answer: A and C

A is the in-place rotation rule. C follows three clockwise quarter turns versus one counterclockwise quarter turn. B ignores relative orientation. D invents a goal-triggered halt. Award one mark only for A and C together.

Q6. Answer: C

C turns north, then moves one row up. A reverses the statement order. B omits the move. D turns the wrong way and moves south.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: After first: (4,2). After second: position (4,2), heading north. Final: (3,2,N).

1 mark: Gives position (4,2) after the first movement.

1 mark: States position remains (4,2) after rotation.

1 mark: Gives heading north after rotation.

1 mark: Gives final state (3,2,N).

Q8. 4 marks

Answer: forward=false; left=true; right=true; all three evaluations leave position and heading unchanged.

1 mark: Gives false for forward.

1 mark: Gives true for left.

1 mark: Gives true for right.

1 mark: States that sensor evaluations do not move or rotate the robot.

Q9. 4 marks

Answer: REPEAT 3 TIMES

```
{  
  MOVE_FORWARD()  
}  
ROTATE_LEFT()  
REPEAT 3 TIMES  
{  
  MOVE_FORWARD()  
}
```

Final state: (1,4,N).

1 mark: Uses a three-iteration block containing MOVE_FORWARD for the initial east segment.

1 mark: Places ROTATE_LEFT after the first segment.

1 mark: Uses a three-iteration forward block after the turn for the north segment.

1 mark: Gives final state (1,4,N).

Q10. 4 marks

Answer: A: no move occurs; the attempted blocked move terminates the program, so no rotation. B: no move occurs because the IF body is skipped; the later rotation executes.

1 mark: States A does not change cells.

1 mark: States A terminates before its rotation.

1 mark: States B skips the movement body because the sensor is false.

1 mark: States B continues and executes ROTATE_RIGHT.

Q11. 4 marks

Answer: The final position and heading agree. A uses 1 turn; B uses 3. A executes 7 movement-or-turn commands and B executes 9, so A uses fewer.

1 mark: States that both final position and heading agree.

1 mark: Gives one turn for A.

1 mark: Gives three turns for B.

1 mark: Compares 7 versus 9 executed movement-or-turn commands and identifies A as fewer.

Q12. 4 marks

Answer: The marker does not stop the program. The next move would leave the grid and is illegal. The robot remains at (1,4), still north. The program terminates, so ROTATE_RIGHT does not execute.

1 mark: States that reaching the marker does not automatically halt execution.

1 mark: Identifies the following move as off-grid and invalid.

1 mark: States that the robot remains at (1,4,N).

1 mark: States that termination prevents the later rotation.